

CHAPTER 4 : MEASUREMENT OF MATTER

Topic:-

Prepare models of various radicals using cardboard, small circular magnets and alladite. From these make models of molecules of various compounds.

Introduction:-

Matter is composed of minute particles called atoms and molecules. During chemical reactions, atoms gain, lose, or share electrons to attain stable electronic configurations similar to noble gases. When an atom or a group of atoms carries a positive or negative electric charge and behaves as a single distinct unit in chemical combinations, it is termed a **Radical**.

Radicals are classified into two major categories: **Basic Radicals (Cations)** which carry positive electrical charges (such as Na^+ , Mg^{2+} , Al^{3+} , NH_4^+), and **Acidic Radicals (Anions)** which carry negative electrical charges (such as Cl^- , O^{2-} , SO_4^{2-} , CO_3^{2-}). The combining capacity of a radical corresponds directly to its numerical charge, known as its **Valency**.

To make the abstract concepts of ionic bonding and chemical formula writing engaging and tangible, we designed and built innovative magnetic 3D models of various radicals using stiff cardboard, small circular neodymium magnets, and Araldite epoxy adhesive. By linking complementary magnetic radicals, we constructed accurate molecular models of diverse compounds.

❖ Key Highlights of Radical Modeling:-

- Opposite magnetic poles mimic electrostatic attraction between oppositely charged ions.
- The number of magnets embedded on a cutout directly indicates the valency of the radical.
- Compounds form only when total positive valencies magnetically balance total negative valencies.

Aim & Setup:-

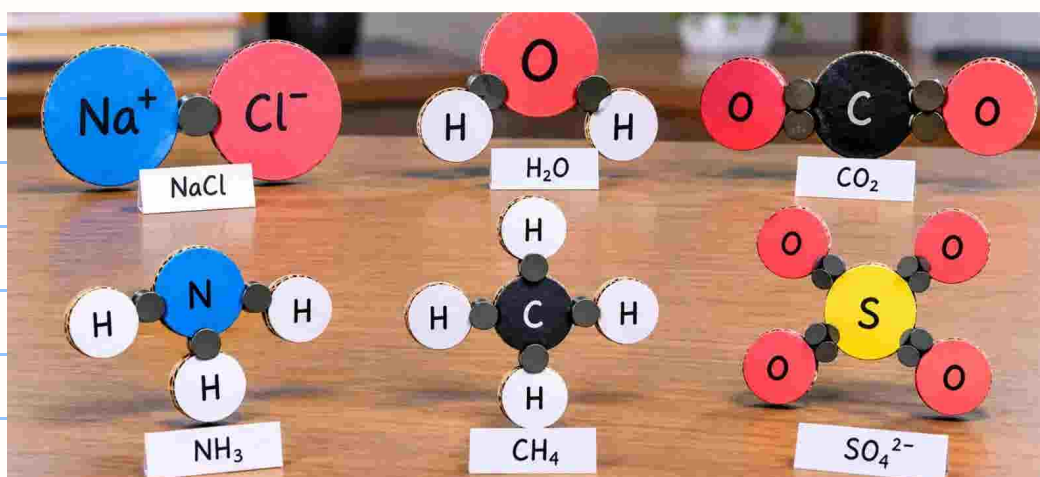
Aim: To construct tangible magnetic models of basic and acidic radicals using cardboard, circular magnets, and Araldite adhesive, and combine them to assemble models of diverse chemical compound molecules.

❖ Objectives of the Project:-

- To design standardized color-coded cardboard templates representing monovalent, divalent, and trivalent radicals.
- To securely bond small circular magnets onto cardboard cutouts using Araldite epoxy resin.
- To assemble molecular models of common compounds (NaCl , MgCl_2 , Al_2O_3 , CaCO_3 , $(\text{NH}_4)_2\text{SO}_4$).
- To verify the criss-cross method of chemical formula formulation through hands-on magnetic assembly.

❖ Materials & Tools Required:-

- Stiff ivory cardboard sheet, scissors, cutter, compass, and acrylic color paints (Blue & Red).
- Small circular neodymium disc magnets (diameter 10 mm, thickness 2 mm; North/South oriented).
- Araldite epoxy adhesive (resin + hardener two-tube adhesive system) and mixing stick.



Step-by-Step Model Construction:-

The construction of magnetic radical models was completed through the following systematic procedure:

❖ *Fabrication & Assembly Procedure:-*

- **Step 1 (Template Design):** Drew uniform geometric shapes on cardboard. Basic radicals (cations) were shaped as interlocking convex puzzle pieces painted blue; acidic radicals (anions) were concave pieces painted red.
- **Step 2 (Labeling):** Labeled each shape clearly with its symbol and charge: Na^+ , K^+ , Mg^{2+} , Ca^{2+} , Al^{3+} , NH_4^+ , Cl^- , O^{2-} , SO_4^{2-} , CO_3^{2-} , PO_4^{3-} .
- **Step 3 (Adhesive Preparation):** Mixed equal portions of Araldite resin and hardener thoroughly on a scrap cardboard strip using a spatula until a uniform sticky paste was formed.
- **Step 4 (Magnetic Embedding):** Firmly glued circular magnets into pre-cut sockets at cutout bonding tabs. For all cations, the North pole faced outward; for all anions, the South pole faced outward. The number of magnets attached matched the radical's valency (1 magnet for valency 1; 2 magnets for valency 2; 3 magnets for valency 3).
- **Step 5 (Curing & Testing):** Allowed the adhesive to cure undisturbed for 12 hours. Tested magnetic snapping strength between complementary radical shapes.

Observations & Radical Classification Table:-

We systematically cataloged our prepared magnetic models, documenting radical type, valency, embedded magnet count, and resulting compound formulas:

Radical Name	Symbol	Classification	Valency	Synthesized Compound Model
Sodium ion	Na^+	Basic (Cation)	1	NaCl (Sodium Chloride)
Magnesium ion	Mg^{2+}	Basic (Cation)	2	MgCl_2 (needs 2 Cl^- models)
Aluminium ion	Al^{3+}	Basic (Cation)	3	Al_2O_3 (2 Al^{3+} snap with 3 O^{2-})
Ammonium ion	NH_4^+	Basic (Composite)	1	$(\text{NH}_4)_2\text{SO}_4$ (with 1 SO_4^{2-})
Chloride ion	Cl^-	Acidic (Anion)	1	CaCl_2 (with 1 Ca^{2+})
Sulphate ion	SO_4^{2-}	Acidic (Composite)	2	CaSO_4 (Calcium Sulphate)
Carbonate ion	CO_3^{2-}	Acidic (Composite)	2	CaCO_3 (Calcium Carbonate)

Every assembled compound demonstrated perfect physical and magnetic equilibrium with zero unbound magnetic poles.

Scientific Analysis & Formula Verification:-

The hands-on assembly of magnetic radical models provides a vivid physical verification of fundamental laws of chemical combination:

❖ Scientific Analysis & Principles Verified:-

- **Verification of the Criss-Cross Method:** When assembling Magnesium Chloride, one Mg^{2+} model has two magnetic poles. Attempting to attach only one Cl^- model leaves one magnet uncoupled. A second Cl^- model must magnetically snap to satisfy the second pole, visually proving why the chemical formula is MgCl_2 and not MgCl !
- **Formation of Complex Molecules (Al_2O_3):** Two Al^{3+} models present a total of 6 positive poles ($2 \times 3 = 6$). Three O^{2-} models present a total of 6 negative poles ($3 \times 2 = 6$). Snapping them together forms a sturdy, fully saturated 3D composite unit representing Al_2O_3 .
- **Composite vs Simple Radicals:** Simple radicals consist of a single charged atom (e.g., Na^+ , Cl^-), whereas composite radicals contain groups of covalently bonded atoms carrying an overall net charge (e.g., NH_4^+ , SO_4^{2-}). The models clearly represented composite radicals as integrated single pieces.

Advantages & Educational Significance:-

Our low-cost magnetic models offer tremendous practical and pedagogical benefits for secondary school chemistry education:

❖ Key Educational Advantages:-

- **Demystifies Chemical Formulas:** Students often struggle with memorizing subscripts in formulas like $\text{Al}_2(\text{SO}_4)_3$. Physical magnetic linking provides immediate tactile and visual clarity.
- **Self-Correcting Learning Tool:** If a student attempts to combine radicals in the wrong ratio, unmatched magnetic poles or awkward dangling ends immediately signal an unbalanced valency error.
- **Eco-Friendly & Inexpensive:** Made entirely from discarded packaging cardboard, scrap magnets, and Araldite, costing less than commercial chemical model kits.

❖ Practical Limitations & Care Tips:-

- Araldite requires adequate drying time; handling before curing can displace magnet polarity.
- Strong neodymium magnets must be kept away from computer screens, credit cards, and mobile memory chips.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Gained deep conceptual understanding of basic radicals, acidic radicals, and variable valencies.
- Mastered the art of translating abstract molecular formulas into tangible 3D interactive models.
- Acquired practical skills in workshop fabrication, accurate polarity alignment, and epoxy adhesion.

❖ Project Conclusion:-

Our magnetic radical models successfully demonstrated the formation of electrically neutral molecules through valency satisfaction. The magnetic bonding accurately models ionic electrostatic attraction between oppositely charged radicals. This project demonstrates that creative, hands-on modeling converts intimidating chemistry concepts into intuitive, joyful, and permanent learning.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 4: Measurement of Matter).
- NCERT Class 9 Science – Atoms and Molecules, Structure of Atom.
- National Science Teachers Association (NSTA) – Innovative Hands-on Models for Teaching Chemical Bonding.